MILITARY SPECIFICATION

CARTRIDGE, IGNITER, MARK 164 MOD 0

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes requirements for manufacture and acceptance of the Cartridge, Igniter, Mark 164 Mod 0, herein referred to as the "igniter cartridge" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following standards of the issue listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods
MIL-STD-1167	Ammunition Data Card

2.1.2 Other Government documents, drawings, and publications. The following drawings, and publications form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions , deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

Naval Air Systems Command (Code Ident 30003)

759AS103 Cartridge, Igniter Assembly, Mark 164 Mod 0

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.2 Design and construction. Unless otherwise specified in the contract (see 6.2.1), the igniter cartridge shall conform to the requirements specified herein and to Drawing 759AS103.

3.3 Environmental requirements. The igniter cartridges shall meet the performance requirements (see 3.5) after being exposed to any of the following conditions or any combination thereof:

a. Temperature cycling (4.4.2.2)
b. Thirty-day storage (4.4.2.3)
c. Vibration (4.4.2.4)

3.4 Radiographic. All loaded igniter cartridges shall be radiographically inspected for the presence of ignition material/propellant (see 4.6.1).

3.5 Ballistic performance requirements. The igniter cartridges shall meet the closed-bomb requirements specified in table I (see 4.6.2).

TABLE I. Closed bomb performance requirements.

Parameter	Temperature (°F) −65	+160
Maximum pressure (lb/sq. in.)		
Minimum	2,800	3,000
Maximum	3,400	3,600
Time to maximum pressure (msec)		
Minimum	20	20
Maximum	30	30
Maximum ignition delay (sec)	0.020	0.020
Secondary maximum pressure (lb/sq. in.)		
Minimum	1,200	900
Maximum	1,900	1,600
Time to secondary maximum pressure (msec) from first pressure indication		
Minimum	0.1	0.1
Maximum	2.0	2.0

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3.6 Service life. The igniter cartridge shall have a minimum service life of 5 years from the date of ignition material/propellant manufacture.

3.7 Workmanship. All components and assemblies shall be free from burrs, contaminations, corrosion, sharp edges, or foreign material which could result in malfunction of the igniter cartridge or components or be a safety hazard in handling.

4.QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1) inspection conditions shall be as specified in the applicable test method of 4.6.

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Sample size. Twenty igniter cartridges shall be submitted for first article inspection.

4.4.2 Inspection routine. First article inspection shall consist of the tests of 4.6 and the additional tests specified in 4.4.2.1 through 4.4.2.4.2 in the sequence specified in table II.

4.4.2.1 Control test firings. Ten igniter cartridges shall be temperature conditioned for 2 hours: five at $-65° \pm 5°F$ and five at $160° \pm 5°F$. The cartridges shall be radiographed (see 4.6.1) and closed-bomb tested at each conditioning temperature in accordance with 4.6.2.

4.4.2.2 Temperature cycling. Four igniter cartridges shall be subjected to temperature cycling. Two cartridges shall be placed in a conditioning chamber which has been preconditioned to $-65° \pm 5°F$. This temperature shall be maintained for 2 hours. The cartridges shall be removed from the chamber and, within a maximum of 5 minutes, shall be placed in a chamber preconditioned to $160° \pm 5°F$. This temperature shall be maintained for 2 hours. The cartridges shall be removed from the conditioning chamber and, within a maximum of 5 minutes, shall be placed in the chamber preconditioned to $-65° \pm 5°F$. This cycling shall be repeated three times. (A cycle is defined as from $-65°$ to $160°$ and return to $-65°F$ or as from $160°$ to $-65°$ and return to $160°F$.) The other two cartridges shall be temperature cycled as above except the temperatures shall be reversed. After cycling the cartridges shall be radiographed (see 4.6.1) and closed-bomb tested (see 4.6.2) at the final cycle temperature.

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TABLE II. First article test schedule.[1/]

Requirement para.	Test para.	Test	Igniter cartridges Control group (cartridges 1 - 10)																			
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
3.3	4.4.2.2	Temperature cycling −65°F, 160°F, −65°F 160°F, −65°F, 160°F											a	a							a	a
3.3	4.4.2.3	Thirty-day storage at −65°F at 160°F													a	a	a	a	a	a		
3.3	4.4.2.4	Vibration at −65°F at 160°F																			b	b
3.4	4.6.1	Radiographic	a	a	a	a	a	a	a	a	a	a	b	b	b	b	b	b	b	b	c	c
3.5	4.6.2	Closed-bomb at −65°F at 160°F	b	b	b	b	b	b	b	b	b	b	c	c	c	c	c	c	c	c	d	d

1/ Letters refer to testing sequences; tolerances for all temperatures shall be ± 5°F.

4.4.2.3 Thirty-day storage. Three igniter cartridges shall be stored for 30 days at 160° ± 5°F and three igniter cartridges at −65° ± 5°F. After 30 days, the cartridges shall be removed from storage, radiographed, conditioned to the temperature at which they were stored, and closed-bomb tested (see 4.6.2).

4.4.2.4 Vibration. Two igniter cartridges shall be subjected to vibration testing. One igniter cartridge shall be conditioned to −65° ± 5°F and one to 160° ± 5°F for 2 hours before testing. The cartridges shall be maintained at these temperatures during the tests. If the igniter cartridge temperature falls outside the specified temperature limits for a period of more than 5 minutes, the testing shall stop and the igniter cartridge shall be reconditioned. A reconditioning time of 2 minutes for every minute the igniter cartridge spent outside the specified temperature envelope shall be required for a maximum of 2 hours. A continuous temperature history of the igniter cartridges shall be kept for all conditioning and test times. The igniter shall be placed in a tube to simulate service use and capped at one end. The opposite end of the tube shall be applied to the holding fixture. Accelerometers to measure acceleration along the axis perpendicular to the axis through the attachment points of the igniter fixture shall be placed on the surface of the tube as close as possible to the end cap. Longitudinal vibration shall be measured at the same location.

4.4.2.4.1 Resonance search. Resonance searches shall be conducted on the igniter cartridges in three axes at the maximum conditioning temperatures. Input vibratory acceleration shall be in accordance with curve H, figure 514-2 of MIL-STD-810. All resonances shall be as noted. During the resonance search a permanent, continuous record of input vibratory acceleration and of unit response to the input vibratory acceleration shall be kept. This record shall be made on FM tape and shall include the response along each of the three axes. Frequency shall be varied at a rate not greater than 0.25 octave per minute.

a. A resonance shall be defined as a frequency at which a clearly defined peak in the amplitude ratio (ratio of the output acceleration in any of the three igniter cartridge axes to the input acceleration as measured at the excitation attachment points) is greater than a value of 1.5 to 1.

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b. If the resonance survey indicates a frequency band wherein the amplitude ratio is greater than 1.5 to 1 (as contrasted to a clearly defined peak) this entire band width shall be considered one resonant point. The minimum frequency band width which shall be considered a resonant band shall be 0.33 octave. If, also within this band width, there are clearly defined amplitude peaks, the most severe of these shall be considered and additional resonant point.

4.4.2.4.2 Endurance vibration. Endurance vibration shall be conducted on the two igniter cartridges at the maximum and minimum conditioning temperatures. Each cartridge shall be vibrated for 24 hours, 8 hours in each of the three axes specified in 4.4.2.4.1. The vibration time in each axis shall be equally divided between the resonant frequencies noted in that axis during the resonant search. Input vibratory acceleration for each resonant frequency shall be in accordance with the acceleration and displacement of curve H, figure 514-2, MIL-STD-810. If no resonant frequencies are detected in an axis, the igniter cartridge shall be vibrated in accordance with curve H, figure 514-2 of MIL-STD-810. Total vibration time for the igniter cartridge shall be 24 hours which shall include resonant and nonresonant frequencies, as applicable. Endurance vibration of frequency bands which are classified as resonances in 4.4.2.4.1 shall be conducted as continuous repetitive sweeps across that entire band up in frequency and return at a rate not greater than 0.25 octave per minute. A record shall be kept on FM tape for a 30-second duration at 30-minute intervals (see 6.2.2). At the conclusion of the tests, the igniter cartridge shall be visually and radiographically inspected for damage without disassembly of the cartridge. The igniter cartridges shall be fired at the same temperature at which they were conditioned and vibrated.

4.4.3 Failures. Failure of any cartridge to pass the first article test shall be cause for first article rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in 4.6.

4.5.1 Lot size. A lot shall consist of not more than 500 igniter cartridges.

4.5.2 Sampling. Sampling shall be as specified in the individual test method of 4.6.

4.5.2.1 Ballistic test. The sample shall be selected by the procuring activity for ballistic testing after lot manufacture has been completed. Igniter cartridge samples shall not be preselected during production. Igniter cartridges selected as samples shall not be applied as part of the quantity specified for delivery by the contract or purchase order. One ignition material production batch may be used in more than one loaded igniter cartridge production lot. More than one ignition material batch may be used. The sample size shall be as equally divided as possible among ignition material batches. The sample shall be selected as follows:

Lot size	Sample size
Up to and including 180	15 igniter cartridges or 2 cartridges per ignition material batch, whichever is greater - 8 cartridges tested at −65°F, 7 cartridges tested at +160°F
Up to and including 500	25 igniter cartridges or 2 cartridges per ignition material batch, whichever is greater - 13 cartridges tested at −65°F, 12 cartridges tested at +160°F

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4.6 Test methods.

4.6.1 Radiographic examination. All igniter cartridges shall be radiographically inspected in accordance with MIL-STD-453, radiographic quality level I at ambient temperature. Any defect shall be cause for rejection of the cartridge. Radiographs shall be capable of detecting defects 0.03 inch in length or width, whichever is smaller. Acceptance of the igniter lot shall be subject to procuring activity approval of the radiographs.

4.6.2 Closed-bomb testing. Igniter cartridges selected in accordance with 4.5.2.1 shall be fired at a test facility designated by the procuring activity within 30 days after submission of the lot. The igniter cartridge shall be placed in the center body of a Rocket Motor Mk 51 Mod 0. The rocket motor chamber (figure 1) shall have all propellant tubes and nozzles removed and the holes welded closed. The igniter cartridge shall be placed in the igniter cartridge tube, and the firing mechanism shall be replaced and torqued to 40 foot-pounds. The free volume shall be 335 ± 10 ml with the cartridge installed. The sample shall be temperature conditioned for a minimum of 2 hours and test-fired within 5 minutes after removal from the conditioning chamber. Any cartridge failing to meet the requirements of 3.5 shall be cause for rejection of the entire lot represented.

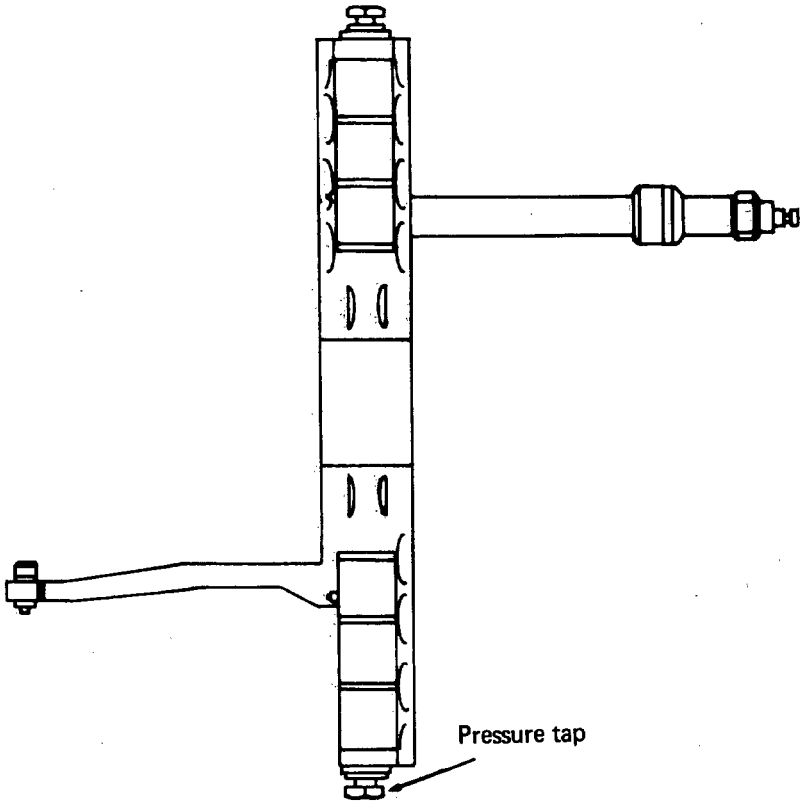


FIGURE 1. Closed-bomb test setup.

4.6.3 Dimensional examination. All igniter cartridges shall be measured for conformance to 3.2 and 3.3. Any cartridge which fails to meet the requirements of 3.2 and 3.3 shall be rejected.

4.6.4 Visual examination. All igniter cartridges shall be visually examined for conformance to 3.2, 3.3, 3.8, and section 5 of this specification. Any cartridge which fails to meet the requirements of 3.2, 3.3, 3.8, and section 5 shall be rejected.

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4.7 Data submission. At the time the igniter cartridges are submitted, the supplier shall include the following information (see 6.2.2):

a. All computations and data required for ballistic evaluation including firing traces
b. Ignition material designation(s), manufacturer, lot number, and date of manufacture
c. Type of test equipment
d. Deviations from standard testing procedure
e. Additional comments (malfunctions, unusual occurences)
f. A summary of test results for parameters given in 3.5 which shall include, but not be limited to, sample size, number and nature of defects found, and disposition (accepted or rejected).

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A.

5.1.1 Cleaning. Each loaded igniter cartridge shall be thouroughly cleaned before packaging.

5.1.2 Packaging. The igniter cartridges shall be packaged in containers subject to the approval of the procuring activity. The package shall be cushioned adequately to protect the igniter cartridges against handling and shipping shocks and vibrations and shall comply with the applicable requirements of MIL-P-116.

5.2 Packing. Packing shall be level A. Igniter cartridges packaged as specified in 5.1.2 shall be packed in an approved container.

5.3 Marking. In addition to any special marking required by the drawings, contract, or purchase order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. Markings shall include, but not be limited to, the following:

a. Manufacturer's name and address
b. Gross weight
c. Lot and serial number
d. Date of propellant manufacture.

5.4 Data cards. Data cards, in accordance with MIL-STD-1167, shall be filled out and included for each production lot. Two cards shall accompany the shipping ticket.

6. NOTES

6.1 Intended use. The igniter cartridges are intended for use as a means of igniting the Underseat Ejection Rocket Motors Mk 51 Mod 1, and Mk 92 Mod 1.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. If first article is required (see 3.1)
c. Design and construction, if other than as specified in 3.2
d. Date and place of delivery

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e. Inspection conditions, if other than as specified in 4.3
f. Special marking, if any (see 5.3)
g. Test facility for closed-bomb testing

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of Federal Acquisition Regulations (FAR) 52-209-3 and -4 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4.2.4.1 and 4.4.2.4.2	Environmental Test Plan/Procedures	DI-R-7125	-
4.7	Inspection and Test Report	DI-T-5329	-
5.4	Ammunition Data Card	DI-L-1410	-

(Data item descriptions related to this specification, and identified in this section will be approved and listed as such in DoD 5000.19L., Vol.II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Maximum pressure. The maximum pressure is defined as the highest point on the pressure-time curve.

6.3.2 Time to maximum pressure. Time to maximum pressure is defined as the elapsed time from the first indication of igniter cartridge pressure to time of maximum pressure on the pressure-time curve.

6.3.3 Secondary maximum pressure. The secondary maximum pressure is defined as the first curve fluctuation measured on the pressure trace after initial indication of cartridge pressure, before continuing to rise to primary maximum pressure. See figure 2.

6.3.4 Ignition delay. Ignition delay is defined as the elapsed time from the removal of sear to first indication of igniter cartridge pressure.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of twenty units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the documents first article.

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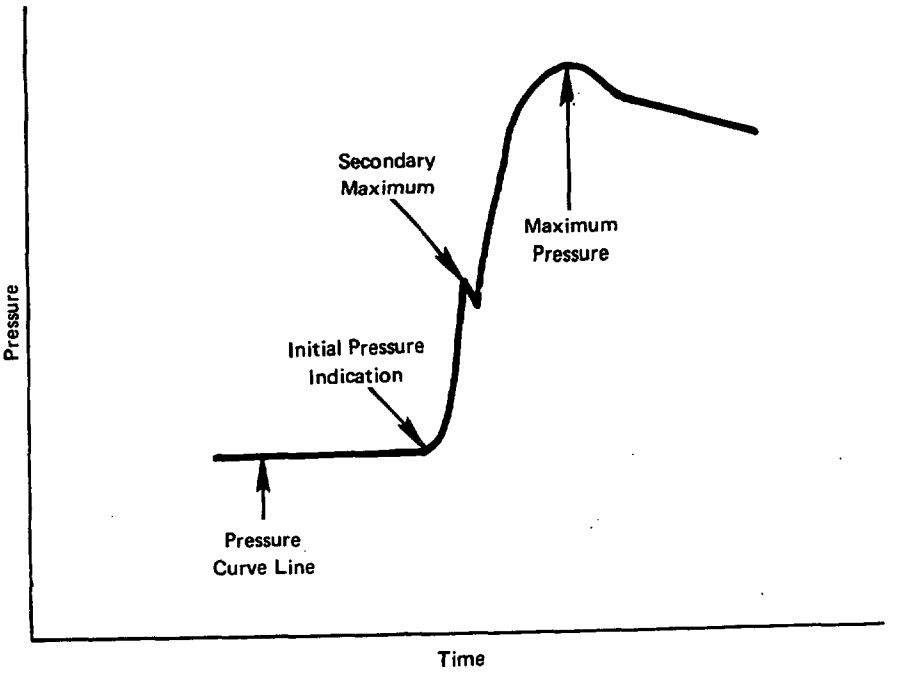


FIGURE 2. Sample pressure tracing.

Preparing activity:
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(Project 1377-N762)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85727(AS)	Cartridge, Igniter, Mark 164 Mod 0

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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